

Introduction

Given the limited time to prepare for Olympiads, we recognise that most people are unable to study every single physics topic before a competition. We have thus given a rough estimate of the weightage for each major topic in the Physics Olympiads, along with bolding important subtopics in the main topic to study. To make the guide more instructive to secondary school students, who may struggle when faced with concepts that are not commensurate with the secondary school level, we have also noted down concepts that are only tested at the JC level; this should help in organizing which topics to study in what order.

Besides the major topics, we have also included a math section first. This is because math and physics are intricately intertwined; how can one solve a problem on Lorentz forces if they are unaware of the cross product? Thus, the math section aims to equip readers with the basic understanding of mathematical concepts, and the knowledge to apply it in a physics-based context. However, please note that we are assuming that a minimum baseline of Sec 3 knowledge of mathematical concepts like algebra and trigonometry. If you are a Sec 2 student or below who is reading these notes, brushing up on your math fundamentals would be more helpful than reading the contents of these notes.

Of course, if you know some of the topics listed here, feel free to skip over chapters as you please. This syllabus guide is not meant to be rigid and inflexible: If you don't have an interest in certain chapters and topics, you can certainly skip them. You don't have to study all the chapters in order to reach your desired level of success in an Olympiad; if studying a tiny bit of the topics in the PRISM notes has helped you attain a level of Honorable Mention, then this guide shall have fulfilled its purpose. Furthermore, this syllabus is made in mind of the most prevalent Olympiad formats in Singapore (SJPO, SPhO and IPhO). If you are preparing for another Olympiad with a different syllabus (and even if you are going for the listed Olympiads!), it may be beneficial to come up with your own study order to master the most important concepts first.

Math

- **Vectors**
 - Expressing vectors in component form
 - Expressing vectors in unit vector form
 - Vector addition and subtract
 - Dot product
 - Cross product
 - Scalar triple product
- Complex numbers (JC)
 - Algebraic notation of complex numbers
 - Trigonometrical form
 - Exponential form
 - Multiplication, division and indices of complex numbers
- Statistics
 - Standard deviation
 - Linearising graphs
 - RMSE
 - Chi-squared test
- **Calculus**
 - Limits
 - Differentiation
 - Chain rule
 - Product rule
 - Quotient rule
 - Integration
 - Bounded integration
 - Integration by substitution (JC)
 - Integration by parts (JC)
 - Graphical interpretation of derivatives and integrals
 - Partial differential equations (JC)
 - Taylor Series (JC)

Topic A: Mechanics (30%)

- **Kinematics**
 - Displacement, velocity, acceleration and their graphical interpretations
 - Free-body diagrams
 - Momentum and impulse in 1D and 2D
 - Thrust equation for rockets
 - Centripetal and tangential acceleration
 - Coriolis effect
- **Equilibrium**
 - Calculation of center of mass
 - Translational and rotational equilibrium conditions
 - Stable and unstable equilibrium
 - Young's modulus
 - Bulk modulus
 - Shear modulus
- **Conservation of energy**
 - Derivation of work equation as an integral
 - Gravitational potential energy
 - Elastic potential energy
 - Static, kinetic and rolling frictional force
 - Kinetic energy
 - Translational kinetic energy
 - Rotational kinetic energy
- **Rotational motion**
 - Moment of inertia
 - Moment of inertia for specified rigid bodies
 - Torque
 - Angular momentum
 - Conservation of angular momentum
 - Parallel axis theorem
 - Derivation of moment of inertia by integration
- **Fluid dynamics**
 - Pressure
 - Hydraulic press systems
 - Buoyancy and resistive forces
 - Laminar and turbulent flow
 - Flow tubes

- Continuity law
 - Bernoulli's equation
- Viscosity
 - Navier-Stokes equation (JC)
- Surface tension
- Capillary pressure
- Gravitational force
 - Newton's Law of Gravity
 - Work done by gravity and gravitational potential energy
 - Escape speed
 - Kepler's Laws
 - Geometry of an ellipse
 - Derivation of second law
 - Derivation of third law
 - Total energy of a system
 - Schwarzschild radius of a black hole (JC)

Topic B: Oscillations, Waves and Light (20%)

- **Simple Harmonic Motion (SHM)**
 - Relationship between force, velocity and displacement
 - Using angular velocity to determine frequency and period
 - Wave phases
 - Using SHM to approximate real and idealised pendulums
 - Critical damping in oscillation (JC)
 - Forced oscillations (JC)
- **Standing and Travelling Waves**
 - Transverse and longitudinal waves
 - Relationship between wave velocity, wavelength and frequency
 - Derivation of travelling wave motion equation using wavenumber
 - Traveling wave speed on a rope with a fixed end
 - Power of a wave
 - Reflection of waves with different boundary conditions
 - Standing wave equation
 - Nodes and antinodes
 - Frequency at the fundamental mode
- **Sound waves**
 - Pressure amplitude of sound waves as a function of bulk modulus
 - Speed of sound waves in different media
 - Intensity dissipation of waves
 - Sound intensity
 - Decibel scale (JC)
 - Sound as a standing wave
 - Wave interference
 - Interference of waves of different frequency
 - Doppler effect (JC)
 - Sound barrier
- **Light**
 - Index of refraction
 - Snell's Law
 - Total internal reflection
 - Critical angle
 - Convex and concave lenses
 - Polarisation of light (JC)
 - Specific rotation order

- Malus' Law
 - Drude's equation for optical rotatory dispersion
 - Brewster angle
- Huygen's Principle (JC)
- Fermat's Principle (JC)
- Interference of light
 - Young's double-slit experiment
 - Bragg's diffraction grating (JC)
 - Michelson interferometer (JC)

Topic C: Electromagnetism (25%)

- **Introduction to Electromagnetism**
 - Relationships between charge, current, voltage, power and resistance
 - Kirschoff's voltage law and Kirschoff's current law
 - Electrostatic forces between charges
 - Faraday's Law
 - Introduction to magnetic flux
 - Biot-Savart Law for simple systems
 - AC and DC introduction
 - Lorentz force
 - Image charges
- Integrals in electromagnetism (JC)
 - Ampere's Law
 - Gauss's Law
 - Superposition principle
- **Advanced electromagnetism**
 - Permeability of free space
 - Dielectric and magnetic permeability (JC)
 - Permittivity of free space
 - Resistivity and conductivity
 - Differential form of Ohm's law (JC)
 - Energy density of an electric field
 - Transformers
 - Step-up transformers
 - Step-down transformers
 - Loss of power due to eddy currents (JC)
 - Charge motion in magnetic fields (JC)
 - Cyclotron frequency
 - Helicoidal motion
- **Circuits**
 - Work done by EMF
 - Joule's Law
 - Circuit components
 - Voltmeter, Ammeter and Ohmmeter
 - Current sources
 - Capacitors and inductors (JC)
 - Self-induction

- Energy of capacitors and inductors
- Time constants of RL and RC circuits
- Impedance of resistors, capacitors, inductors and combination circuits
- Resonance of current and voltage
- Advanced AC circuitry (JC)
 - Usage of complex numbers to quantify AC
 - Phasor diagrams
 - Active power

Topic D: Thermodynamics (15%)

- **Introduction to Thermodynamics**
 - Laws of Thermodynamics
 - Thermal equilibrium
 - Entropy (JC)
 - Work in relation to pressure and volume
 - Difference between open, closed and isolated systems
 - Kinetic molecular theory
 - Boltzmann distribution curve
 - Boltzmann constant
 - Internal energy of ideal gas systems
 - Root Mean Square velocity of a particle
 - Gas laws
 - Ideal gas law and deviation from reality
 - Van der Waals gas law (JC)
 - Redlich-Kwong gas law (JC)
 - Heat engines (JC)
 - Isothermal, isobaric, isochoric and adiabatic processes
 - Carnot cycle
 - Stirling cycle
 - Efficiency of different heat engines
- **Heat transfer and phase transitions**
 - Latent heat capacity
 - Phase transitions
 - Dalton's Law
 - Saturated vapor pressure (JC)
 - Relative humidity (JC)
 - Heat flux (JC)
 - Continuity equation
- **Advanced thermodynamics (JC)**
 - Planck's Law
 - Wien's displacement law
 - Stefan-Boltzmann law

Topic E: Quantum Physics (10%)

- **Structure of matter**
 - Introduction to emission spectrums
 - Neutrinos and quarks (JC)
 - Introduction to fermions (JC)
 - Pauli exclusion principle for protons, neutrons and electrons
 - Compton scattering (JC)
 - Atomic nuclei
 - Strong nuclear force and weak nuclear force
 - Mass defect
 - Radioactivity
 - Alpha, beta and gamma particles
 - Radioactive decay and half-life
 - Fission, fusion and neutron capture
- Relativity (JC)
 - Principle of relativity
 - Lorentz transformation
 - Mass-energy equivalence
 - Invariance of the space-time interval
 - Time dilation
 - Relativistic Doppler effect
 - Lorentz contraction
 - Relativity of simultaneity
 - Energy and momentum of photons
 - Relativistic equation of motion
 - Relativistic collisions
- Probability waves (JC)
 - Wave-particle duality
 - Relationship between frequency and energy
 - Relationship between wave vector and momentum
 - Energy levels of hydrogen-like atoms
 - Hydrogen emission spectra
 - Spectral width and lifetime of excited state hydrogen atoms
 - Parabolic potential well
 - Quantization of angular momentum
 - Uncertainty principle
 - Time and energy uncertainty

- Coordinate and momentum uncertainty